

STUDENT ACTIVITY HANDOUT

IFY STATISTICS | SEMESTER 1 - WEEK 2B

Sampling and reliable results



Name _____ Class _____ Date _____

CLASS NOTES

Calculator allowed. Show your working and explain your choices. Give percentages and money to **2 decimal places**, unless a task asks otherwise. Exercise numbers match the PPT.

REVISION

Quick revision

Slides 3-4

A college has **3,216 students**. All **268 randomly selected students** reply. **173** support longer library hours.

B-R1 Identify the population and the sample.

POPULATION AND SAMPLE

B-R2 Calculate the support percentage. Is it a parameter or a statistic? Explain.

CALCULATION, TYPE AND REASON

B-R3 Name each method:
(a) Random distinct IDs from the full list.
(b) A random start followed by every 12th student.

(a) AND (b)

B-R4 A separate list has **2,592 people**. Select **96** systematically. Find the interval and the first four positions from random start **17**.

INTERVAL CALCULATION AND FIRST FOUR POSITIONS

STRATIFIED SAMPLING: CALCULATE AND SELECT

EXERCISE B1

A proportional sample

Slides 12-13

A company has **2,478 employees**. Select **177** using proportional stratified sampling. Calculate the allocation for each department. Show each calculation.

Task / department	Employees	Calculation	Number selected
B1.1 Production	1,386	 	
B1.2 Sales	714	 	
B1.3 Administration	378	 	

Check that your allocations total 177: _____

B1.4 Describe an exact random selection process within each department.

SELECTION PROCEDURE

EXERCISE B1

The budget changes

Slides 14-15

B1.5 The same company can now select only **173 employees**. Calculate each exact allocation, then choose whole numbers that keep the total at 173.

Department	Exact allocation*	Rounded down	Final allocation
Production	 	 	
Sales	 	 	
Administration	 	 	

*Keep the fraction or show at least four decimal places before allocating whole people.

Explain which departments receive the extra places after rounding down. Check your final total.

ROUNDING DECISION AND TOTAL CHECK

CLUSTER AND STRATIFIED SAMPLING: COMPARE THE PLANS

EXERCISE B2

Three selection plans

Slides 20-21

A different college has **54 classes of 42 students**. Each student belongs to one class. Complete lists are available.

For each plan, **name the method**, **find the sample size** and **explain your classification**.

B2.1 Randomly choose 7 distinct students from every class.

Method:	Sample size / calculation:
Reason:	

B2.2 Randomly choose 9 distinct classes and invite everyone.

Method:	Sample size / calculation:
Reason:	

B2.3 Randomly choose 378 distinct IDs from the college list.

Method:	Sample size / calculation:
Reason:	

EXERCISE B2

Is the selection valid?

Slides 22-23

Use the same 54 classes. For each plan, identify the fault, explain who is favoured and give a precise repair.

B2.4 Take the first 7 names from every class register.

FAULT, WHO IS FAVOURED, AND REPAIR

B2.5 Visit the 9 classrooms nearest the staff office and invite everyone in them.

FAULT, WHO IS FAVOURED, AND REPAIR

SAMPLING VARIATION, BIAS AND MEASUREMENT

EXERCISE B3

Which denominator?

Slides 29-30

A college randomly invites **263 students**. Of these, **157 reply** and **112 respondents support** longer library hours.

B3.1 Calculate the response rate.

B3.2 Calculate support among respondents.

CALCULATION AND ANSWER

CALCULATION AND ANSWER

B3.3 Can we claim the same support across the college? Explain exactly when nonresponse could distort this result.

CONCLUSION AND CONDITION FOR BIAS

EXERCISE B3

Diagnose and repair

Slides 33-34

For each case, identify the **main issue**, its **possible effect** and a **repair**. If a fault is not established, explain that.

Case	Main issue	Possible effect	Repair / response
B3.4 Survey evening-transport needs only in 9 a.m. classes.			
B3.5 In a random staff survey, dissatisfied employees are twice as likely to reply.			
B3.6 A tutor asks named students: "Our feedback system is excellent, isn't it?"			
B3.7 A laboratory balance overstates every reading by 0.18 g.			
B3.8 Two valid random samples estimate travel time at 27.4 and 29.1 minutes.			

ACCURACY AND PRECISION: READ THE GRAPH

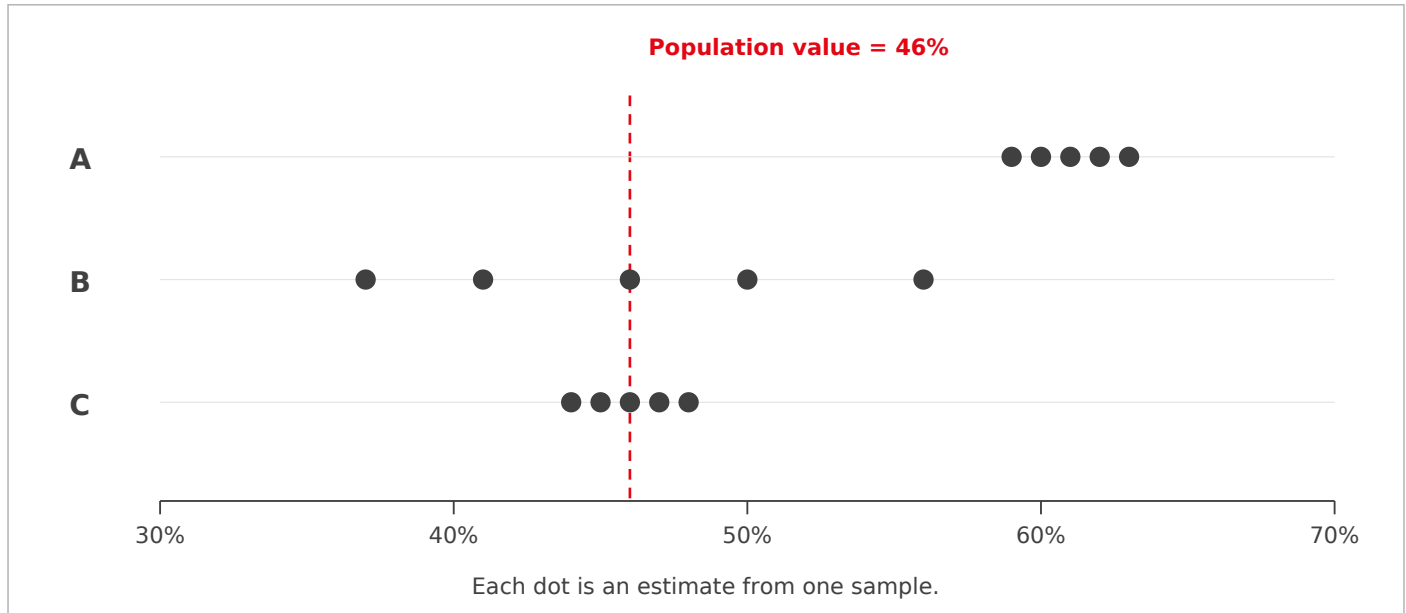
EXERCISE B4

Accuracy and precision

Slides 39-40

Accuracy: closeness to the truth. **Precision:** consistency across samples.

B4.1 Which processes are precise? Which centre near the truth? Explain why consistency alone is not enough.



(a) Which processes are precise? Use the spread of their dots as evidence.

PROCESSES AND EVIDENCE

(b) Which processes centre near the population value? Use the graph as evidence.

PROCESSES AND EVIDENCE

(c) Why is consistency alone not enough to make a process trustworthy?

EXPLANATION

CHOOSE AND DEFEND A SAMPLING PLAN

EXERCISE B4

A sampling decision

Slides 42-43

Estimate support for evening classes. There are **72 centres**, each with **34 trainees**. Select **238**. Complete lists are available. Assume everyone takes part.

Plan A	Plan B
Randomly select 238 individual IDs . This draw covers 67 centres .	Randomly select 7 whole centres . Interview all 34 trainees in each.

Cost: ¥85 per centre visited + ¥12.50 per trainee. **Budget:** ¥9,000.

B4.2 Name the methods. Calculate each cost and check the budget.

Plan	Sampling method	Cost calculation and total	Within budget?
A	 	 	
B	 	 	

B4.3 Choose a plan. Give a reason and its strongest statistical objection.

MY CHOICE, REASON AND STATISTICAL OBJECTION

B4.4 What extra information would help you decide? Explain how it would help.

INFORMATION NEEDED AND WHY

EVALUATE THE EVIDENCE BEHIND A HEADLINE

EXERCISE B5

Does the headline hold up?

Slides 44-45

A college publishes this survey claim:

COLLEGE NEWS

"79.2% of students back longer library hours"

B5.1 Identify four missing details. Rank the two most important and explain why they matter.

	Missing detail
1	
2	
3	
4	

Most important detail: no. ____ Why?

REASON

Second most important detail: no. ____ Why?

REASON

B5.2 Assume **79.2% is the percentage among respondents**. Rewrite the headline.

MY HEADLINE

Does one decimal place make the claim trustworthy? Explain.

EXPLANATION

INDIVIDUAL EXIT CHECK AND OPTIONAL CHALLENGE

EXERCISE B6

Check your understanding

Slides 46-47

Work independently. Give reasons; method names alone are not enough.

- B6.1** Two departments contain **184** and **327 people**. Researchers randomly select **19 from each**. Name the method. Is it proportional?

METHOD, PROPORTIONALITY AND REASON

- B6.2** A random sample receives **287 invitations**, but only **169 reply**. What type of bias is a concern, and when would it occur?

TYPE OF BIAS AND WHEN IT OCCURS

- B6.3** A team doubles its sample but keeps a leading question. Has it solved the problem? Explain.

ANSWER AND REASON

OPTIONAL B3.9

How much could be missing?

Slides 49-50

Return to the **263 invitees: 157 replied**, and **112 supported** longer library hours. There are **106 nonrespondents**.

- B3.9** Among all 263 invitees, find the smallest and largest possible support percentages under these two scenarios:

(a) Every missing answer is **Oppose**.

(b) Every missing answer is **Support**.

CALCULATION AND ANSWER

CALCULATION AND ANSWER

Could we conclude that a majority of the invitees support the change? Explain.

CONCLUSION ABOUT THE 263 INVITEES
